

DS504/DSL504

Natural Language Processing

EndSem

Time 3 Hrs

1.

(a) Show the steps of transition-based dependency parsing for the phrase “We saw him with binoculars”.

(b) In transition-based dependency parsing, for the left-arc transition, we use the condition that the top of stack should not have a head already, while no such condition is used for right-arc transition. Why is that so? [5+3]

2.

(a) How does LSTM attempt to solve the vanishing gradient problem of RNNs?

(b) What are the different stopping criteria used for beam search?

(c) Why does beam search have the issue of “longer hypotheses have lower scores”? How is it solved? [3+3+3]

3.

Suppose in a Neural Machine Translation problem, the vector corresponding to the first decoder hidden state is [0.3, 0.1] whereas the vectors corresponding to the encoder hidden states are [0.4, 0.1], [0.2, 0.8], [0.5, 0.6] (encoder has three states only). Calculate the attention vector for the first decoder hidden state. [5]

4.

(a) In Transformers, what is the need of scaling in dot product while calculation of attention scores even after adding layer normalization?

(b) Show with an example text, the need for using multiple heads while calculating self attention in transformers.

(c) What are the differences between the attention modules of an encoder and a decoder in a transformer? Justify the difference. [2.5+2.5+3]

5.

(a) Which task is less open-ended between the Dialogue system and Machine translation system? Justify the reasons.

(b) What is the exposure bias issue of decoders? Provide two possible solutions to it.

(c) Briefly discuss a possible way how the retriever used in the RAG setup can be contextualized with respect to the generator (specify the kind of loss function you would use)? [2+4+4]

6.

(a) Give an example of an anaphoric relation which is not coreferential.

(b) Give an example of a coreferential relation which is not anaphoric.

(c) Give an example of a coreferential relation which cannot be resolved correctly using POS tag based methods such as Hobbs'. [2+2+2]

7.

In the Gold Coin system ("GOLD COIN: Grounding Large Language Models in Privacy Laws via Contextual Integrity Theory"), the authors have used a "consistency filter" step while performing case postprocessing. Explain the step and why it is required. [4]

8.

For the SyncTOD system ("Synergizing In-context Learning with Hints for End-to-end Task-oriented Dialog Systems"), explain the purpose and functionality of the hint prediction step with an example. [5]

9.

In the Backward Lens paper ("Backward Lens: Projecting Language Model Gradients into the Vocabulary Space"), the authors have shown a heatmap to explain the norm of GPT2-xl's FF2's VJPs ($\delta \mathbf{v}$) as a function of the layers' index and segments of the edited prompts. Explain what is VJP and what is the observation from this heatmap? [3+2]